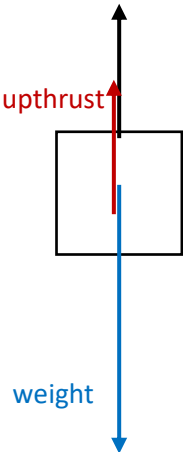


2022 Preliminary Examination Paper 2 Suggested Solution

Qn 1	Answer	Marks
(a) (i)	When mass M is placed on the spring, the spring extends by a length of $(L_2 - L_1)$. $mg = k(L_2 - L_1)$ $k = \frac{mg}{L_2 - L_1}$ $= \frac{(0.0985)(9.81)}{0.037 - 0.013} = 40.3 \text{ N m}^{-1}$	C1 M1
(a) (ii)	$\frac{\Delta k}{k} = \frac{\Delta m}{m} + \frac{\Delta(L_2 - L_1)}{L_2 - L_1} = \frac{0.2}{98.5} + \frac{0.2 + 0.1}{3.7 - 1.3} = 0.127$ Thus $\Delta k = k \times \frac{\Delta k}{k} = 40.3 \left(\frac{0.2}{98.5} + \frac{0.2 + 0.1}{3.7 - 1.3} \right) = 5 \text{ N m}^{-1}$ (Do not penalise if Δk is left to a value that is greater than 1 s.f. – this mark will be marked in (a)(iii))	M1 A1
(a)(iii)	$k = (40 \pm 5) \text{ N m}^{-1}$ Δk expressed to 1 s.f. and k expressed to the same place value as Δk. Allow ecf from (a)(ii) and (a)(iii).	A1
(b)(i)	Tension  upthrust weight Marking Guidance: B1 – 3 forces drawn with correct direction B1 – Relative lengths of forces show $T+U = W$ [-1] – Missing labels/legend Note that first B1 mark must be obtained before the second B1 mark can be given. [Point of action is not marked here. But in general, students should know that the upthrust is acting at the centre of buoyancy – c.g. of the liquid displaced and weight is at the c.g. of the cube.]	B2
(b)(ii)	Initially (Fig. 1.2), P is inequilibrium, net force on it is zero. Weight = upthrust $W = (0.7V)\rho g$	M1
(b)(iii)	After cube Q was connected (Fig. 1.3), at new equilibrium, net force on P is zero. Weight of P = Tension due to string + new upthrust	

	$W = T + U' = T + 0.4V\rho g$ Solving the two equations above gives $T = W - 0.4V\rho g = W - \frac{0.4W}{0.7} = \frac{3}{7}W \text{ (Shown)}$	M1 M1
(b)(iii)	Consider the forces acting on cube Q. Resolving the weight of cube Q along the slope, and since it is in equilibrium, $T = (3/7) W = W \sin (\theta) \Rightarrow \text{Solving, } \theta = 25.4^\circ$	A1
	[Total : 11 marks]	